

Jackson Alvarez

Software Engineer & Automation Architect

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EXECUTIVE SUMMARY

Systems-minded software engineer and automation specialist with experience building scalable production systems, enterprise tooling, and cloud-native workflows across higher education, SaaS, and government-adjacent environments. Proven ability to operate across engineering, infrastructure, security, automation, and executive communication layers simultaneously. Experienced leading technical initiatives involving AWS architecture, compliance readiness, product delivery, deployment pipelines, and operational reporting for high-impact organizations and rapidly evolving products.

PROFESSIONAL EXPERIENCE

Product Support Engineer / Security Engineer

Sept 2025 – Present

Everblue, Hybrid – Charlotte, NC

- Maintained and expanded a production-grade government contracting web application supporting large-scale federal operational workflows and reporting pipelines.
- Worked directly with executive leadership to streamline SOC-2 audit preparation, improve internal operational transparency, and strengthen platform security posture.
- Contributed across the full software lifecycle using Next.js, AWS infrastructure, GitHub workflows, and VSCode-driven development pipelines in a tenant-heavy SaaS architecture.
- Implemented automation workflows, runtime monitoring systems, and reporting infrastructure providing week-over-week and month-over-month operational visibility across enterprise contract activity.
- Supported production deployments, debugging, feature iteration, and internal tooling improvements while collaborating across engineering and business stakeholders.
- Acted as a cross-functional technical resource bridging software engineering, cloud infrastructure, compliance readiness, and security-oriented operational practices.

Founder & Lead Automation Engineer

Mar 2025 – Present

Vector Labs, Charlotte, NC

- Founded an independent software and automation laboratory focused on helping businesses modernize operational workflows, internal systems, and AI-assisted automation infrastructure.
- Architected and prototyped multiple software products involving automation pipelines, AI-integrated tooling, cloud-native infrastructure, and scalable web application systems.
- Designed technical solutions spanning backend engineering, deployment architecture, workflow orchestration, and product strategy for evolving client and internal initiatives.
- Operated across executive planning, engineering leadership, systems design, product iteration, and rapid development environments while managing independent technical direction.
- Focused on transitioning advanced automation concepts into commercially viable products and operational systems for modern businesses.

Systems Engineer I

Mar 2023 – May 2025

Appalachian State University, Boone, NC

- Developed PowerShell automation systems supporting SCCM and Azure AD management across 8,000+ university devices.
- Maintained and optimized 400+ software deployment packages, reducing operational inconsistencies and improving deployment reliability.
- Automated reporting, monitoring, and internal systems workflows supporting enterprise-scale IT operations.
- Collaborated with vendors, engineering teams, and technical staff using Agile methodologies and structured project workflows.
- Authored technical documentation and trained junior developers and student engineers on automation tooling and deployment processes.

Computer Science Intern

May 2022 – Aug 2022

Hypermill.digital, Cornelius, NC

- Built centralized data systems using GraphCRM and Retool to manage 10,000+ product SKUs within a live enterprise data ecosystem.
- Designed scalable software workflows supporting marketing, operational, and support automation initiatives.
- Collaborated with data-oriented teams to structure and manage dynamic product datasets for enterprise usage.

SELECTED PROJECTS

VibeCodeCLI



Cross-language developer CLI assistant supporting execution, testing, and automation workflows across 10+ programming languages.

- Developed dynamic token budgeting, runtime orchestration, and context-aware cost optimization systems.

Runoff Forecasting with LSTM



Machine learning flood prediction system leveraging recurrent neural network architectures and time-series forecasting.

- Preprocessed environmental datasets and trained LSTM models to forecast runoff patterns and hydrological behavior.

EDUCATION

B.S. Computer Science, Minor in Mathematics

July 2025

Appalachian State University, Boone, NC

CORE COMPETENCIES

Programming:

Python, TypeScript, JavaScript, Java, C/C++, SQL, Rust, PHP, Haskell, MATLAB, PowerShell, Lua, R

Technologies:

AWS, Next.js, React, Node.js, Azure AD, GitHub, SCCM, Terraform, TensorFlow, Flask, OpenAI APIs

Leadership Areas:

Systems Architecture, Automation Strategy, Cloud Infrastructure, Security Operations, Product Engineering, Technical Leadership

ADDITIONAL HIGHLIGHTS

- Experience operating in fast-moving startup-style engineering environments with executive-level communication exposure.
- Strong understanding of scalable SaaS architecture, cloud deployment workflows, compliance-oriented engineering, and automation-first operational design.
- Actively building independent software ventures and automation-focused business systems through Vector Labs.
- Interested in executive technology leadership opportunities involving AI systems, infrastructure, automation, and advanced engineering strategy.